

Slowest Marble Record Sheet — Day 1

Name: Dean

1. Identify the Problem

Restate the engineering problem in your own words. What does your structure need to accomplish?

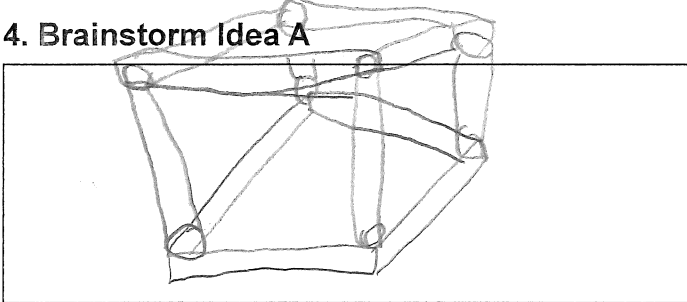
the marble needs to take as long as possible to move from start to end of structure

2-3. What Should We Think About?

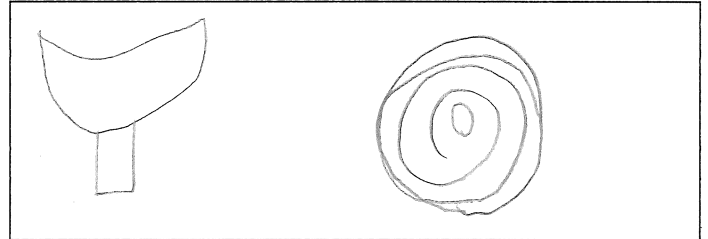
Before building, write or sketch a few things you think could affect travel time and reliable motion. What could help the marble keep moving? What could cause a run to fail?

the more down a slide design the faster it will go.
turning might fail if the marble stops moving

4. Brainstorm Idea A



4. Brainstorm Idea B

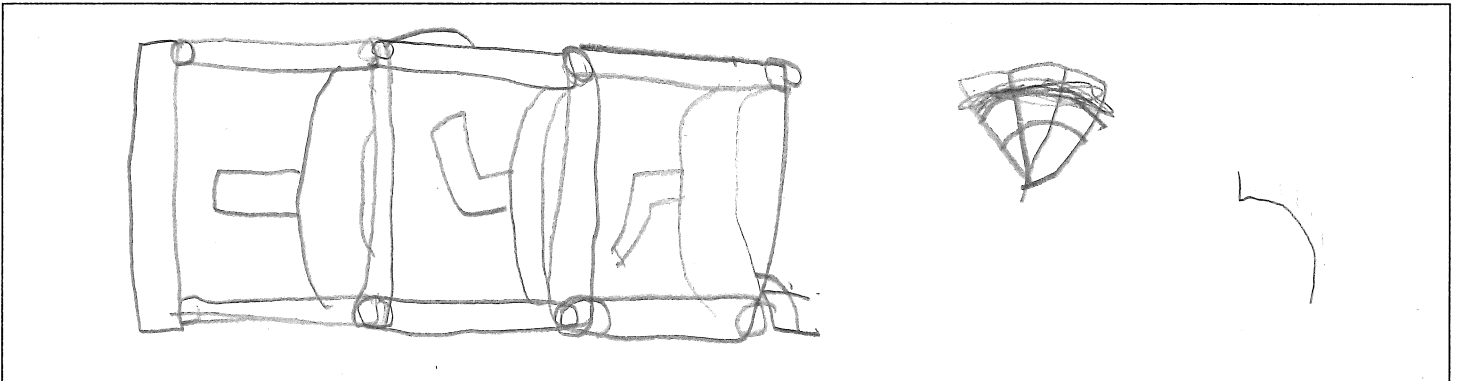


5. Choose a Solution

Which idea will you build first, and why do you think it is the better starting point?

A if the foundation this can be 3 feet high it needs to stay

6. Prototype Plan — Final Sketch Before Building



Slowest Marble — Days 2–3

Name: Pear

7. Version 1 Test — Day 2

Trial	Travel time	Successful?	What happened?
1	2.5	Yes	first design was too complicated
2	1.5	Yes	second design worked
3	None	No	not enough time

Version 1 score — longest successful time: trial 2.5

Evidence check: Where was the marble fastest? Where was it slowest? Where did the run nearly fail?

trial 1, trial 2, nearly failed when small marble fell

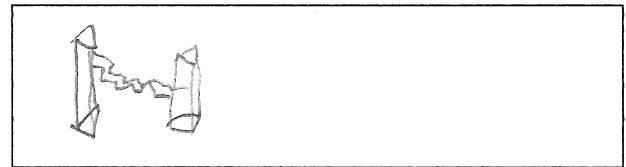
8. Make It Better

Choose one or two specific changes based on Version 1 evidence. Explain what you will change and why you think it will help.

Changes I will make and why:

Supports from square to triangle better
Support from swirl to stairs
Swirl to complicated

Improvement sketch:



7. Version 2 Test — Day 3

Trial	Travel time	Successful?	What happened?
1	1.5	Yes	went down stairs to top
2	1.85	Yes	almost got stuck
3	None	No	No time

Version 2 score — longest successful time: 1.85

Improvement: Version 2 score - Version 1 score = 1.85 - 2.5 = -0.65

9. What Did You Learn?

1. What feature of your final structure had the greatest effect on travel time? What evidence supports your answer?

stairs its smallest time is evidence

3. What was the most important problem you discovered in Version 1?

too complicated

5. What did another team do differently that you think was effective?

more simple designs

2. Why do you think the marble was faster in some parts of the run and slower in others?

slow is where gravity was stronger

4. What change did you make for Version 2? Did it produce the result you expected?

changed supports and stairs
it was faster on the stairs

6. If you had one more redesign, what would you change next and why?

make it simpler so it would

Final Comparison

	Version 1	Version 2
Best successful time	2	1.8
Most effective feature	swirl	stairs
Main weakness / problem	complicated	nothing

Slowest Marble — Days 2–3

Name: itali walter

7. Version 1 Test — Day 2

Trial	Travel time	Successful?	What happened?
1	None	No	kept getting stuck
2	3 seconds	Not really	Also kept getting stuck
3	5 seconds	Kind of	Also kept getting stuck. I think I had to work

Version 1 score — longest successful time: 5 seconds

Evidence check: Where was the marble fastest? Where was it slowest? Where did the run nearly fail?

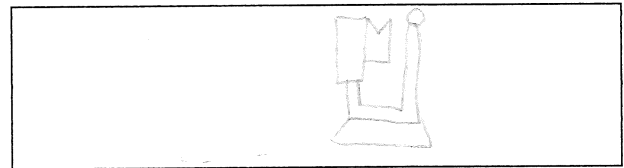
8. Make It Better

Choose one or two specific changes based on Version 1 evidence. Explain what you will change and why you think it will help.

Changes I will make and why:

The first design didn't work
it was too short.

Improvement sketch:



7. Version 2 Test — Day 3

Trial	Travel time	Successful?	What happened?
1	5 sec	Yes	Didn't get stuck
2	5 sec	Yes	Didn't get stuck
3	5 sec	Yes	No problems

Version 2 score — longest successful time: 5 seconds

Improvement: Version 2 score – Version 1 score = None

9. What Did You Learn?

1. What feature of your final structure had the greatest effect on travel time? What evidence supports your answer?

The drop really gets
it going.

3. What was the most important problem you discovered in Version 1?

Re-support was the
placement of things

5. What did another team do differently that you think was effective?

Add support that it's
enough.

2. Why do you think the marble was faster in some parts of the run and slower in others?

Because of the shape. If it
wasn't be slow until the bottom.

4. What change did you make for Version 2? Did it produce the result you expected?

Yes. I think I had to add more
support.

6. If you had one more redesign, what would you change next and why?

To make it longer

Final Comparison

	Version 1	Version 2
Best successful time	5 sec	5 sec
Most effective feature	the drop	Drop
Main weakness / problem	getting stuck at first	getting stuck

Slowest Marble Record Sheet — Day 1

9-9-2026

Name: Middle school

1. Identify the Problem

Restate the engineering problem in your own words. What does your structure need to accomplish?

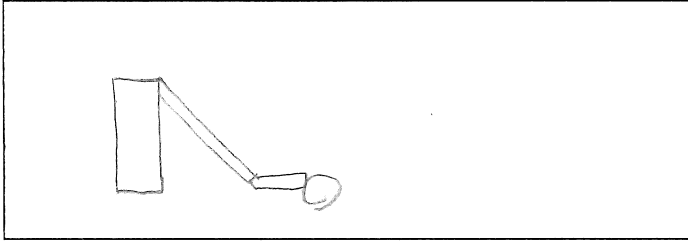
It'll need to reach its destination without falling, without pushing it, and also needs be able to stay up.

2-3. What Should We Think About?

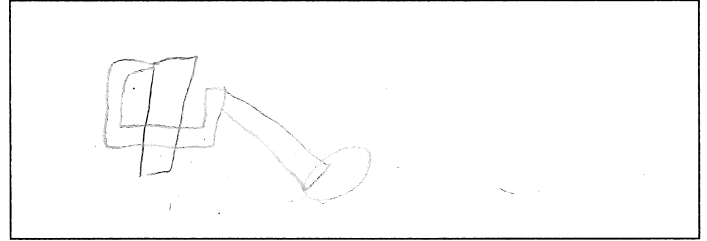
Before building, write or sketch a few things you think could affect travel time and reliable motion. What could help the marble keep moving? What could cause a run to fail?

Hills, bumps, just getting enough speed

4. Brainstorm Idea A



4. Brainstorm Idea B

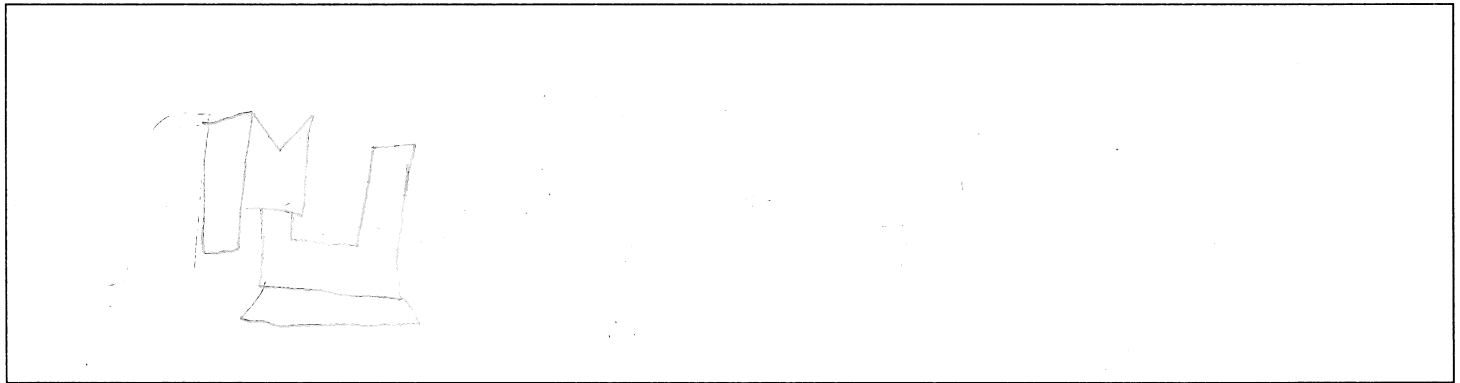


5. Choose a Solution

Which idea will you build first, and why do you think it is the better starting point?

I think brainstorm idea B because its a slide with bumps and easier to make, making it more efficient.
or a tower with a slide

6. Prototype Plan — Final Sketch Before Building



Slowest Marble Record Sheet — Day 1

Name: Chase D

1. Identify the Problem

Restate the engineering problem in your own words. What does your structure need to accomplish?

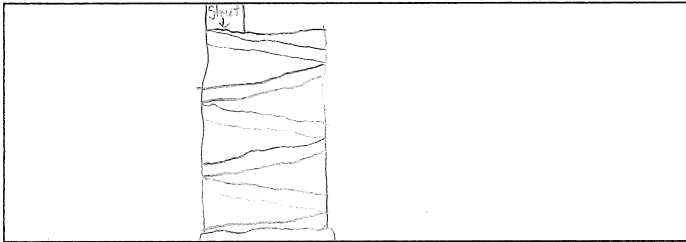
It's needs to free standing structure that make the marble slow and on a long track/longest but slowest rollercoaster

2-3. What Should We Think About?

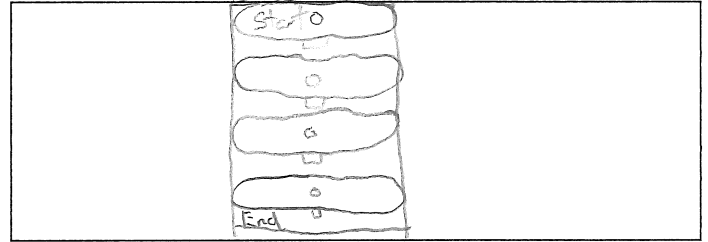
Before building, write or sketch a few things you think could affect travel time and reliable motion. What could help the marble keep moving? What could cause a run to fail?

have stops so the marble keeps moving. what can effect the run is if theres smooth surface so the ball can stay in its path. Something that could cause a run to fail is lik if the ball falls of the track or build. doesn't meet requirement

4. Brainstorm Idea A



4. Brainstorm Idea B

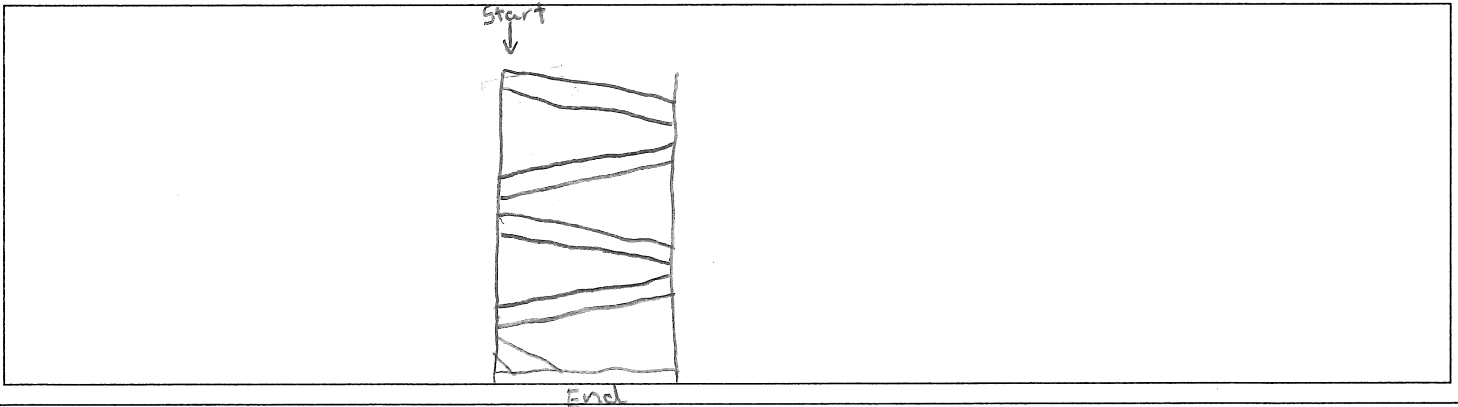


5. Choose a Solution

Which idea will you build first, and why do you think it is the better starting point?

B because gravity won't let it drop straight down the middle it will go around till it reaches the middle.

6. Prototype Plan — Final Sketch Before Building



Slowest Marble — Days 2–3

Name: Chase D

7. Version 1 Test — Day 2

Trial	Travel time	Successful?	What happened?
1	08.44	Yes	Smooth and a little fast
2	09.68	Yes	got stuck on tape a little
3	00.00 —	No	got stuck on dropper

Version 1 score — longest successful time: 9.68

Evidence check: Where was the marble fastest? Where was it slowest? Where did the run nearly fail?

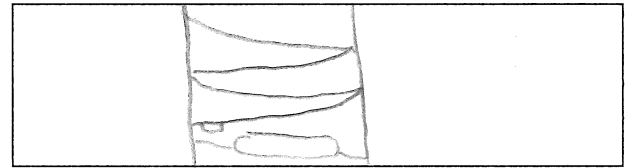
8. Make It Better

Choose one or two specific changes based on Version 1 evidence. Explain what you will change and why you think it will help.

Changes I will make and why:

make track smoother so
Don't get stuck on tape

Improvement sketch:



7. Version 2 Test — Day 3

Trial	Travel time	Successful?	What happened?
1	06.05	Yes	Track is smoother so its faster
2	00.00	Fail	Fell off side when hit side
3	08.80	Yes	

Version 2 score — longest successful time: 08.80

Improvement: Version 2 score – Version 1 score = 0.12

9. What Did You Learn?

1. What feature of your final structure had the greatest effect on travel time? What evidence supports your answer?

When it got stuck a little
Version 1 trial 2

3. What was the most important problem you discovered in Version 1?

The rough tape supports

5. What did another team do differently that you think was effective?

2. Why do you think the marble was faster in some parts of the run and slower in others?

The tape layout on track

4. What change did you make for Version 2? Did it produce the result you expected?

The track more smooth

6. If you had one more redesign, what would you change next and why?

Better tape make it more sturdy

Final Comparison

	Version 1	Version 2
Best successful time	09.68	08.80
Most effective feature	The dropper	The dropper
Main weakness / problem	tape	tape

Slowest Marble Record Sheet — Day 1

Name: Kai

1. Identify the Problem

- also how to stop the friction ~~from~~ from stopping the marble ~~from reaction the finish line~~

Restate the engineering problem in your own words. What does your structure need to accomplish?

How do we keep the marble on ~~the path~~ the path to the finish line.
Also how to keep the structure ~~free-standing~~ free-standing.

- The friction
- losing momentum
- too much speed.

2-3. What Should We Think About?

Before building, write or sketch a few things you think could affect travel time and reliable motion. What could help the marble keep moving? What could cause a run to fail?

How to keep it moving and traveling

- Flatten the slopes
- Use gentle obstacles
- Maximize distance
- Add controlled friction

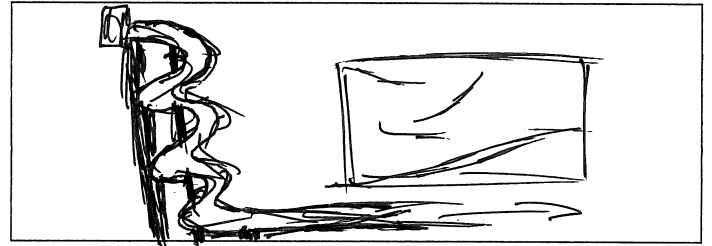
What can cause it to fail

- Loss of momentum / getting stuck
- Jumping the track
- Too much kinetic energy early on
- Poor alignment

4. Brainstorm Idea A



4. Brainstorm Idea B

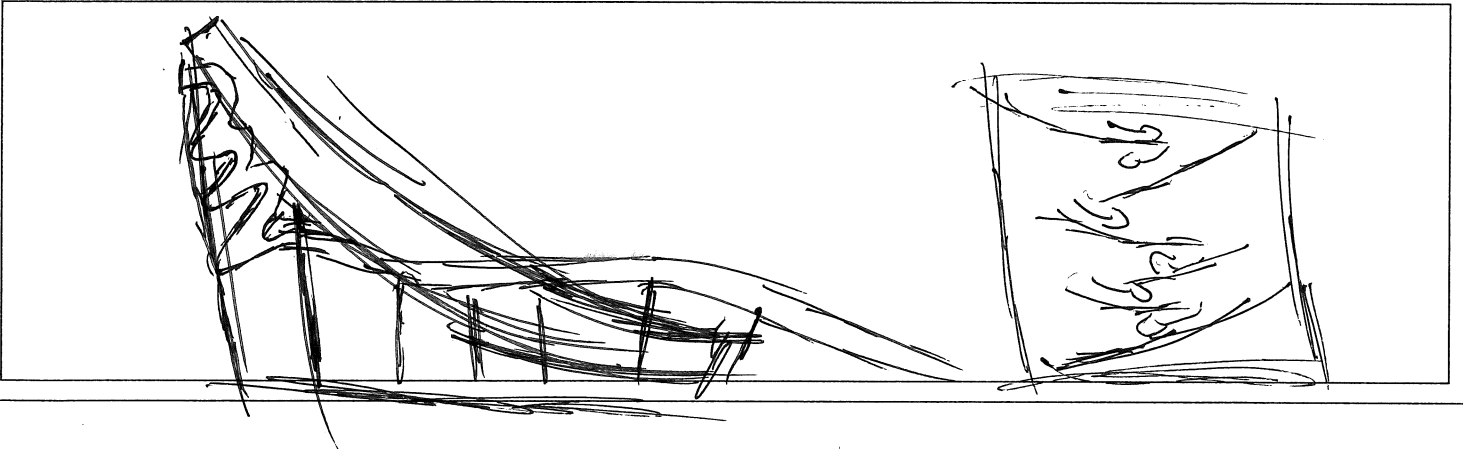


5. Choose a Solution

Which idea will you build first, and why do you think it is the better starting point?

I'm going to go with Idea A because I think it's going take the ~~longest~~ ^{shortest} time to build and I ~~also~~ think it's going to do the best.

6. Prototype Plan — Final Sketch Before Building



Slowest Marble — Days 2–3

Name: Kai

7. Version 1 Test — Day 2

Trial	Travel time	Successful?	What happened?
1	4.86	yes	it went back and forth
2	4.46	yes	
3			

Version 1 score — longest successful time: 4.86

Evidence check: Where was the marble fastest? Where was it slowest? Where did the run nearly fail?

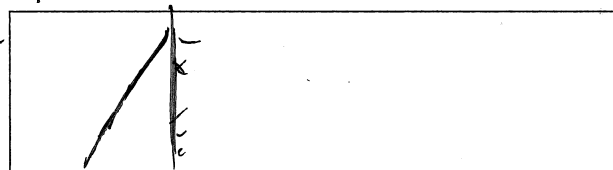
8. Make It Better

Choose one or two specific changes based on Version 1 evidence. Explain what you will change and why you think it will help.

Changes I will make and why:

lean it back more for the marble to stay on the course.

Improvement sketch:



7. Version 2 Test — Day 3

Trial	Travel time	Successful?	What happened?
1	4.86	yes	
2			
3			

Version 2 score — longest successful time: 4.86

Improvement: Version 2 score – Version 1 score = 4.86

9. What Did You Learn?

1. What feature of your final structure had the greatest effect on travel time? What evidence supports your answer?

the first ramp

2. Why do you think the marble was faster in some parts of the run and slower in others?

to big of a ramp

3. What was the most important problem you discovered in Version 1?

how it was standing

4. What change did you make for Version 2? Did it produce the result you expected?

lean in back more

5. What did another team do differently that you think was effective?

make the ramps less downwards

6. If you had one more redesign, what would you change next and why?

how I build it

Final Comparison

	Version 1	Version 2
Best successful time	4.86	
Most effective feature	the ramps	
Main weakness / problem	how it standing	

Slowest Marble Record Sheet — Day 1

Name: Jackman Sharp

1. Identify the Problem

Restate the engineering problem in your own words. What does your structure need to accomplish?

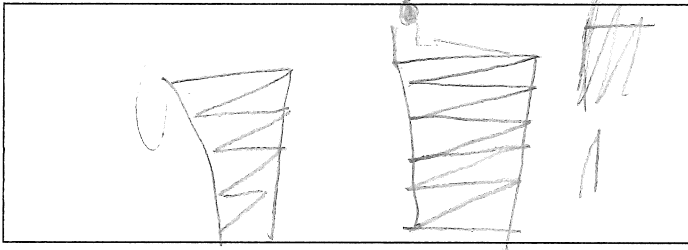
The slowest marble - longest time spent on track moving without finish

2-3. What Should We Think About?

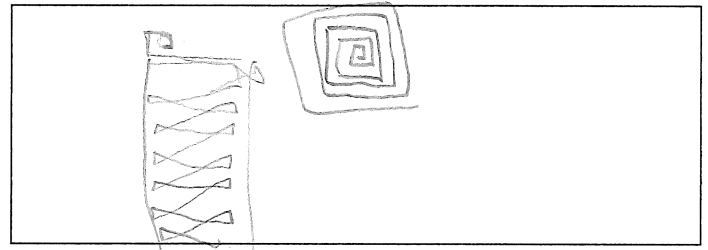
Before building, write or sketch a few things you think could affect travel time and reliable motion. What could help the marble keep moving? What could cause a run to fail?

Not steep / Tall / Within 12 in x 12 in / Max height 36 in

4. Brainstorm Idea A



4. Brainstorm Idea B

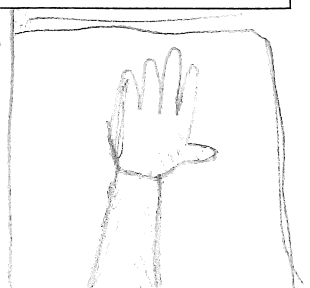
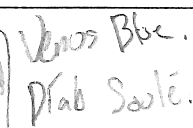
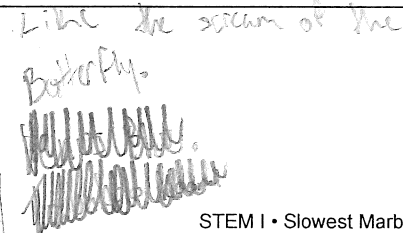
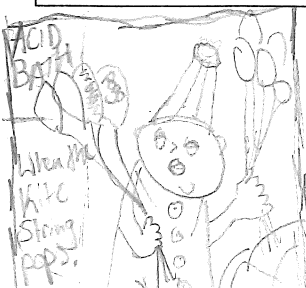
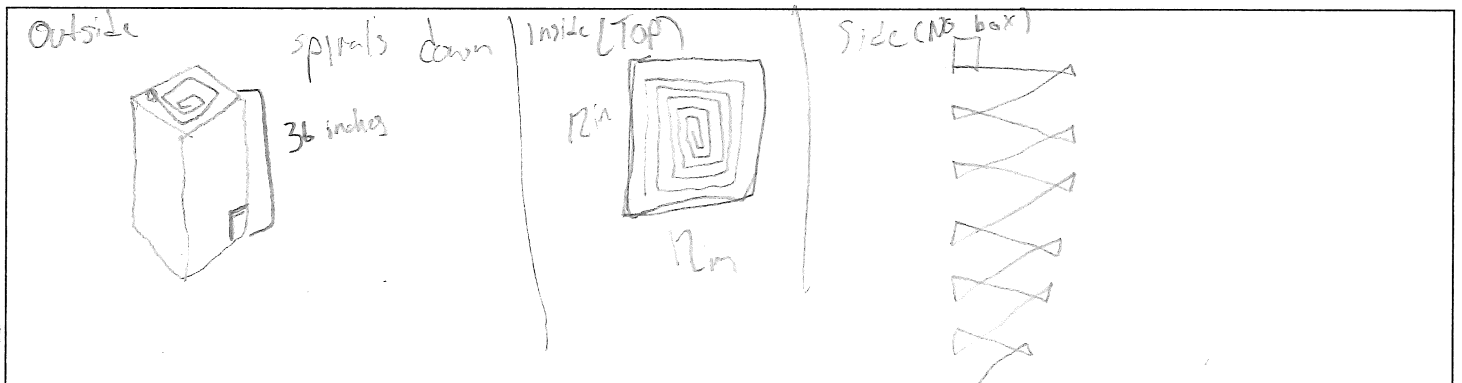


5. Choose a Solution

Which idea will you build first, and why do you think it is the better starting point?

B, put a box around it and have one piece inside

6. Prototype Plan — Final Sketch Before Building



Slowest Marble — Days 2–3

Name: _____

7. Version 1 Test — Day 2

Trial	Travel time	Successful?	What happened?
1	0	no	Fell apart
2	0	no	Fell apart
3	2	no	Fell apart

Version 1 score — longest successful time: 2

Evidence check: Where was the marble fastest? Where was it slowest? Where did the run nearly fail?

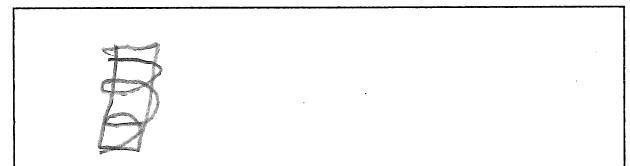
8. Make It Better

Choose one or two specific changes based on Version 1 evidence. Explain what you will change and why you think it will help.

Changes I will make and why:

Stronger Standoff

Improvement sketch:



7. Version 2 Test — Day 3

Trial	Travel time	Successful?	What happened?
1	2	no	Fell apart
2	4	no	Fell apart
3	0	no	Fell apart

Version 2 score — longest successful time: 4

Improvement: Version 2 score – Version 1 score = 2

9. What Did You Learn?

1. What feature of your final structure had the greatest effect on travel time? What evidence supports your answer?

Stronger tape placement

3. What was the most important problem you discovered in Version 1?

Weak

5. What did another team do differently that you think was effective?

Bigger pieces of card board

2. Why do you think the marble was faster in some parts of the run and slower in others?

4. What change did you make for Version 2? Did it produce the result you expected?

6. If you had one more redesign, what would you change next and why?

Final Comparison

	Version 1	Version 2
Best successful time	2	4
Most effective feature	Stronger tape placement	Card board size
Main weakness / problem	Card board size	Not enough tape

Slowest Marble Record Sheet — Day 1

Name: Nelson Bond

1. Identify the Problem

Restate the engineering problem in your own words. What does your structure need to accomplish?

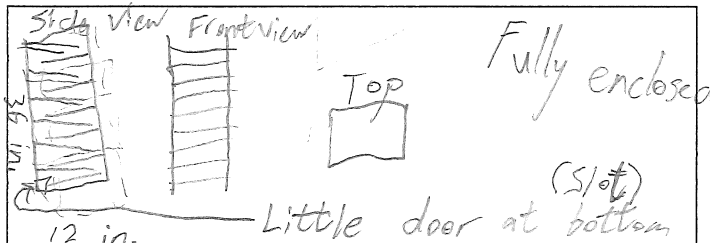
I need to make a cardboard structure that makes a marble take a long time to reach the ground.

2-3. What Should We Think About?

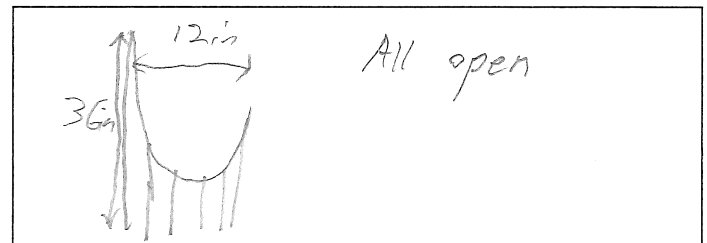
Before building, write or sketch a few things you think could affect travel time and reliable motion. What could help the marble keep moving? What could cause a run to fail?

- Very small slopes will hinder speed
- Cut slots, like a cabin, instead of using tape

4. Brainstorm Idea A



4. Brainstorm Idea B

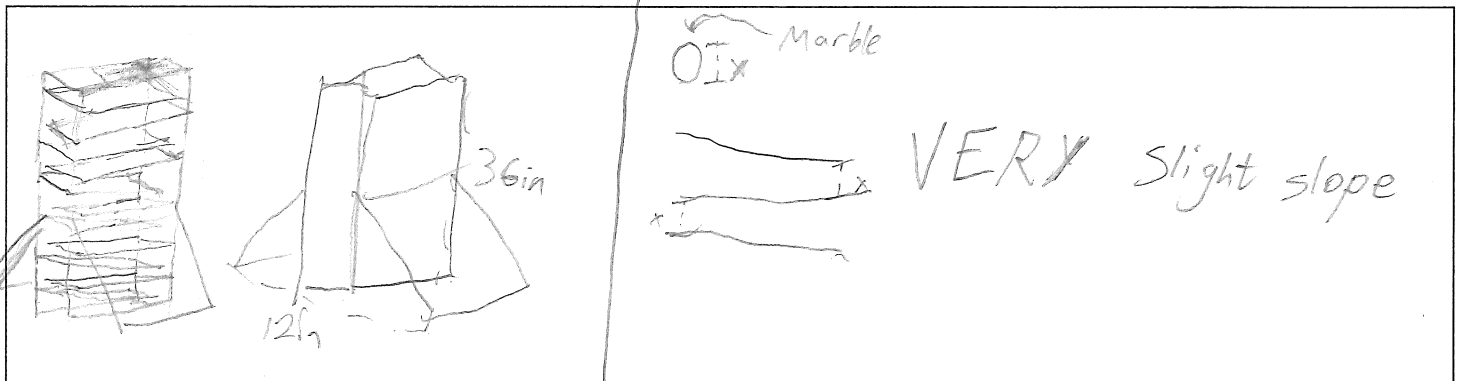


5. Choose a Solution

Which idea will you build first, and why do you think it is the better starting point?

Idea A is what I'll go with first, I think it'll make the marble go slowest, and will also have a sturdier build

6. Prototype Plan — Final Sketch Before Building



Slowest Marble — Days 2–3

Name: _____

7. Version 1 Test — Day 2

Trial	Travel time	Successful?	What happened?
1			
2			
3			

Version 1 score — longest successful time: _____

Evidence check: Where was the marble fastest? Where was it slowest? Where did the run nearly fail?

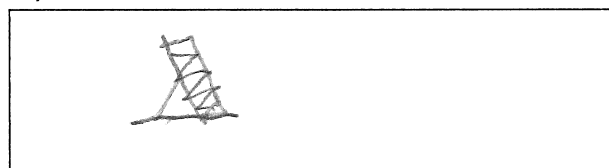
8. Make It Better

Choose one or two specific changes based on Version 1 evidence. Explain what you will change and why you think it will help.

Changes I will make and why:

- Simpler (one side open)
+ Less time building

Improvement sketch:



7. Version 2 Test — Day 3

Trial	Travel time	Successful?	What happened?
1			
2			
3			

Version 2 score — longest successful time: _____

Improvement: Version 2 score - Version 1 score = _____

9. What Did You Learn?

1. What feature of your final structure had the greatest effect on travel time? What evidence supports your answer?

Height / gravity

3. What was the most important problem you discovered in Version 1?

Slope Support

5. What did another team do differently that you think was effective?

Smaller overall structure

2. Why do you think the marble was faster in some parts of the run and slower in others?

Measurements not exact

4. What change did you make for Version 2? Did it produce the result you expected?

Used 7 inch gaps all around, definitely helped

6. If you had one more redesign, what would you change next and why?

Smaller overall structure

Final Comparison

	Version 1	Version 2
Best successful time		
Most effective feature	Slab setup	
Main weakness / problem	Size	

Slowest Marble — Days 2–3

Name: Landen

7. Version 1 Test — Day 2

Trial	Travel time	Successful?	What happened?
1	7 sec	yes	It went down with no problem
2	7:19		
3	7:38		

Version 1 score — longest successful time: _____

Evidence check: Where was the marble fastest? Where was it slowest? Where did the run nearly fail?

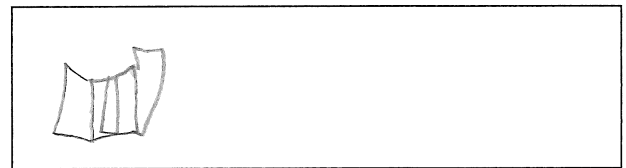
8. Make It Better

Choose one or two specific changes based on Version 1 evidence. Explain what you will change and why you think it will help.

Changes I will make and why:

The marble seems to fall off the rails, I should lean it back

Improvement sketch:



7. Version 2 Test — Day 3

Trial	Travel time	Successful?	What happened?
1			
2			
3			

Version 2 score — longest successful time: _____

Improvement: Version 2 score – Version 1 score = _____

9. What Did You Learn?

1. What feature of your final structure had the greatest effect on travel time? What evidence supports your answer?

Leaning it back

2. Why do you think the marble was faster in some parts of the run and slower in others?

more of a tilt

3. What was the most important problem you discovered in Version 1?

for it to go slow you need to lean it back

4. What change did you make for Version 2? Did it produce the result you expected?

I made a stand for my structure and leaned it back

5. What did another team do differently that you think was effective?

better tilt on my rails

6. If you had one more redesign, what would you change next and why?

Final Comparison

	Version 1	Version 2
Best successful time	7:38	
Most effective feature	Stand	
Main weakness / problem	It's fast	

Slowest Marble Record Sheet — Day 1

Name: hansen

1. Identify the Problem

Restate the engineering problem in your own words. What does your structure need to accomplish?

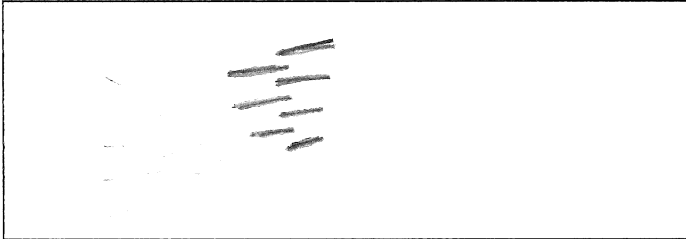
Design and build a free standing cardboard structure that makes a marble take as long as possible to travel from the designated start to the finish using gravity.

2-3. What Should We Think About?

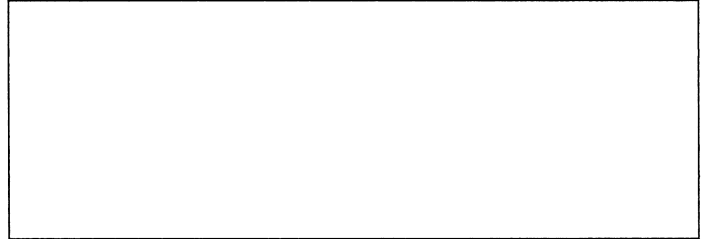
Before building, write or sketch a few things you think could affect travel time and reliable motion. What could help the marble keep moving? What could cause a run to fail?

The angle of what I do, it needs to be a very slight angle.

4. Brainstorm Idea A



4. Brainstorm Idea B

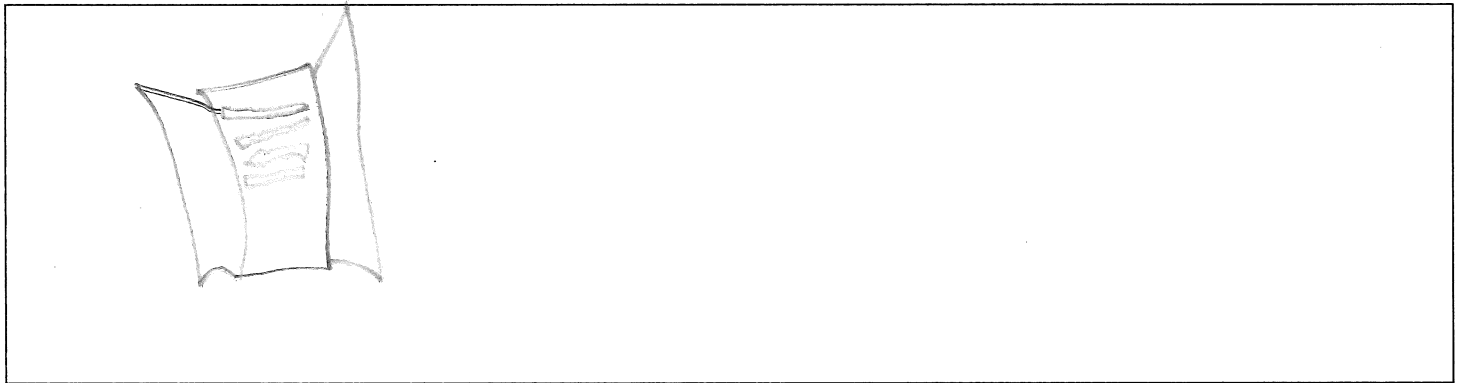


5. Choose a Solution

Which idea will you build first, and why do you think it is the better starting point?

I will do A

6. Prototype Plan — Final Sketch Before Building



Slowest Marble Mini Challenge

Engineering Design Process in Action • Challenge Sheet

Name: _____ Class: _____ Date: _____

The Challenge

Design and build a free-standing cardboard structure that makes a marble take as long as possible to travel from the designated start to the finish using only gravity.

Material Budget

- Unlimited cardboard
- Maximum **48 inches of masking tape** in the final structure
- One standard teacher-provided marble for official tests
- No other materials may be part of the final structure

Cardboard may be reused during redesign.

Requirements

- Entire structure stays within a **12 in × 12 in base area**.
- Maximum height is **36 inches**.
- Structure must be free-standing.
- After release, the marble moves only because of gravity.
- No touching, blowing, shaking, tilting, or other interaction after release.
- The marble must stay on the designed path, keep moving, and reach the finish.

Testing Procedure

1. Verify that the completed structure meets the base, height, material, and tape limits.
2. Place the teacher-provided marble at the designated starting point.
3. Release the marble without giving it a push.
4. Time the run from release until the marble crosses the designated finish.
5. A trial counts only if the marble moves continuously, stays on the designed path, remains untouched, and reaches the finish.
6. Complete **3 official trials**. Record the **longest successful time** as the version score.
7. Use what you learned to improve the design, then repeat the same test for Version 2.

Success Measure

Primary: Longest successful travel time. **Also track:** Improvement from Version 1 to Version 2.

Engineering Design Process

- | | |
|---|--|
| ☐ Identify the Problem | ☐ Research |
| ☐ Requirements | ☐ Brainstorm Ideas |
| ☐ Choose a Solution | ☐ Build a Prototype |
| ☐ Test It Out | ☐ Make It Better |
| ☐ Share Results | |

Long Span Record Sheet — Day 1

Name: Hannah

1. Identify the Problem

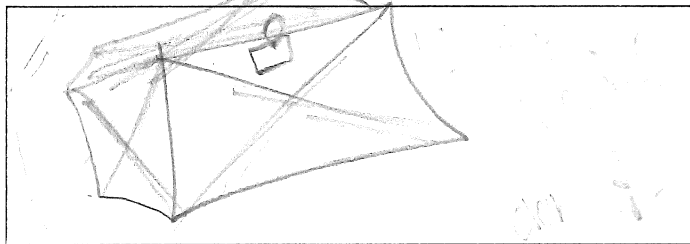
Restate the engineering problem in your own words. What does your structure need to accomplish?

Create the greatest clear supported span

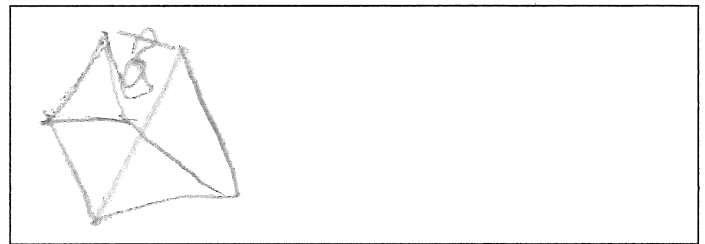
2–3. What Should We Think About?

Before building, write or sketch a few things you think could affect span and strength. Consider shape, joints, bending, bracing, and how the hanging load will travel through the structure.

4. Brainstorm Idea A



4. Brainstorm Idea B

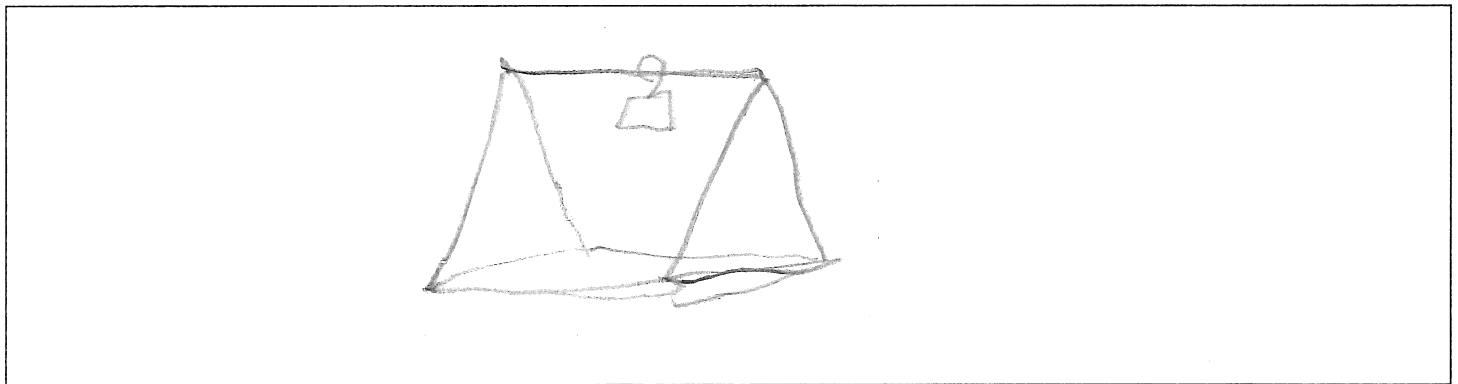


5. Choose a Solution

Which idea will you build first, and why do you think it is the better starting point?

A, but B would be better

6. Prototype Plan — Final Sketch Before Building



7. Version 1 Test

Maximum successful span	Straws used	Toothpicks used	Tape used
2 straws ?	24	0	20 in

What happened first as the structure began to fail?

didn't fail, it held fine.

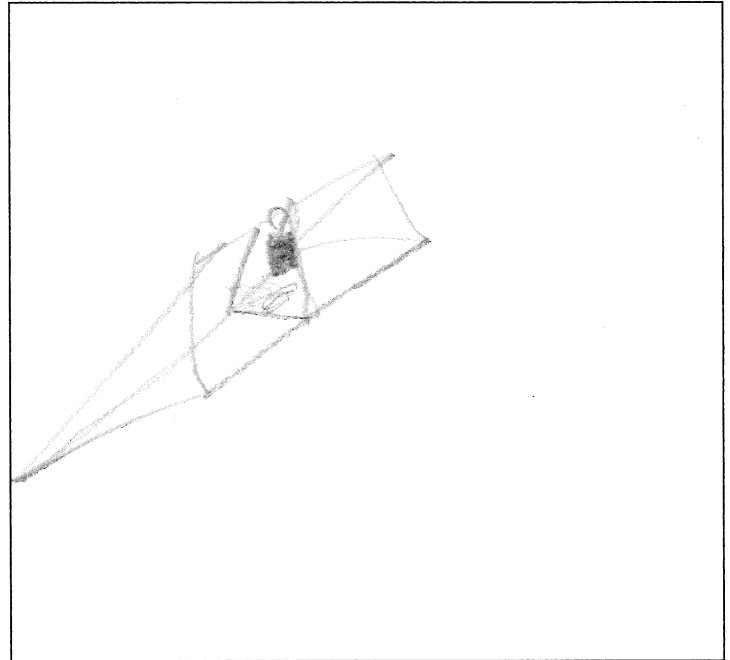
8. Make It Better

Choose one or two specific changes based on the first test. Explain what you will change and why you think it will help.

Changes I will make and why:

add straws to the bottom

Improvement sketch:



7. Version 2 Test

Maximum successful span	Straws used	Toothpicks used	Tape used

9. What Did You Learn?

Did Version 2 improve? What evidence supports your answer? If you had one more redesign, what would you change next?

Final Comparison

	Version 1	Version 2
Clear span		
Best feature		
Main weakness		

Slowest Marble Record Sheet — Day 1

Name: Cheyenne Martin

1. Identify the Problem

Restate the engineering problem in your own words. What does your structure need to accomplish?

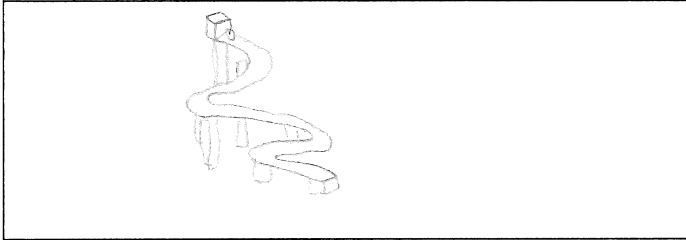
Design and build a free-standing cardboard structure that uses only gravity to make a marble travel from the start to the finish as slowly as possible.

2-3. What Should We Think About?

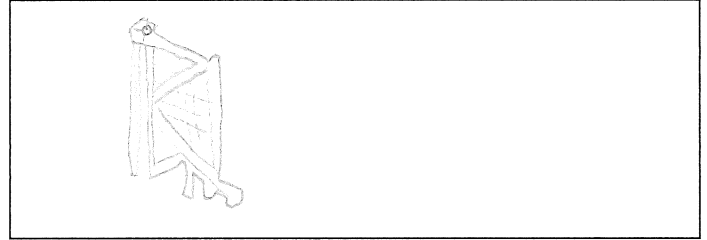
Before building, write or sketch a few things you think could affect travel time and reliable motion. What could help the marble keep moving? What could cause a run to fail?

- Track Slope - slight downward slope
- Friction - too much = stop not enough = move too quickly
- Track Shape - curves, turns, and longer pathways

4. Brainstorm Idea A



4. Brainstorm Idea B

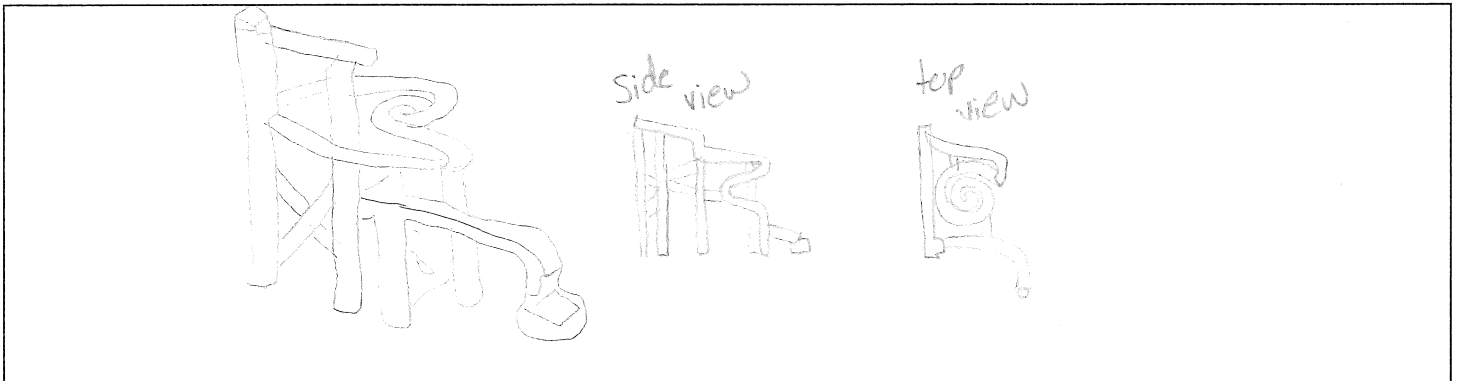


5. Choose a Solution

Which idea will you build first, and why do you think it is the better starting point?

Idea A because it can make the marble travel a longer distance and take more time.

6. Prototype Plan — Final Sketch Before Building



Slowest Marble — Days 2–3

Absent

Name: _____

7. Version 1 Test — Day 2

Trial	Travel time	Successful?	What happened?
1			
2			
3			

Version 1 score — longest successful time: _____

Evidence check: Where was the marble fastest? Where was it slowest? Where did the run nearly fail?

8. Make It Better

Choose one or two specific changes based on Version 1 evidence. Explain what you will change and why you think it will help.

Changes I will make and why:

Improvement sketch:

7. Version 2 Test — Day 3

Trial	Travel time	Successful?	What happened?
1			
2			
3			

Version 2 score — longest successful time: _____

Improvement: Version 2 score – Version 1 score = _____

9. What Did You Learn?

1. What feature of your final structure had the greatest effect on travel time? What evidence supports your answer?
2. Why do you think the marble was faster in some parts of the run and slower in others?
3. What was the most important problem you discovered in Version 1?
4. What change did you make for Version 2? Did it produce the result you expected?
5. What did another team do differently that you think was effective?
6. If you had one more redesign, what would you change next and why?

Final Comparison

	Version 1	Version 2
Best successful time		
Most effective feature		
Main weakness / problem		

Slowest Marble — Days 2–3

Name: Adam Olson

7. Version 1 Test — Day 2

Trial	Travel time	Successful?	What happened?
1	5 sec	NO	Ball rolled off the sides
2	10 sec	yes	fixed everything
3	12 sec	yes	✓

Version 1 score — longest successful time: 12 sec

Evidence check: Where was the marble fastest? Where was it slowest? Where did the run nearly fail?

Marble was fastest at the start, slowest at the end.

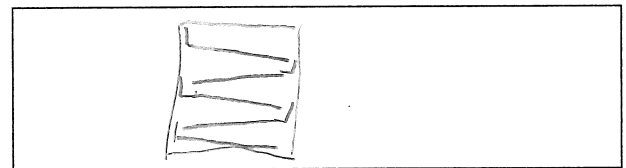
8. Make It Better

Choose one or two specific changes based on Version 1 evidence. Explain what you will change and why you think it will help.

Changes I will make and why:

walls so Ball doesn't go off the side

Improvement sketch:



7. Version 2 Test — Day 3

Trial	Travel time	Successful?	What happened?
1	15 sec	✓	Everything worked!
2	15 sec	✓	
3	15 sec	✓	

Version 2 score — longest successful time: 15

Improvement: Version 2 score - Version 1 score = 15 - 12 = 3

9. What Did You Learn?

1. What feature of your final structure had the greatest effect on travel time? What evidence supports your answer?

walls on the side so it didn't fall

3. What was the most important problem you discovered in Version 1?

It kept falling off the sides

5. What did another team do differently that you think was effective?

made track longer

2. Why do you think the marble was faster in some parts of the run and slower in others?

Because of the slant

4. What change did you make for Version 2? Did it produce the result you expected?

added walls

6. If you had one more redesign, what would you change next and why?

add more tape for the sides

Final Comparison

	Version 1	Version 2
Best successful time	12 sec	15 sec
Most effective feature	down slope	walls
Main weakness / problem	Ball stopped	walls breaking

Slowest Marble Record Sheet — Day 1

Name: Adam Olsen

1. Identify the Problem

Restate the engineering problem in your own words. What does your structure need to accomplish?

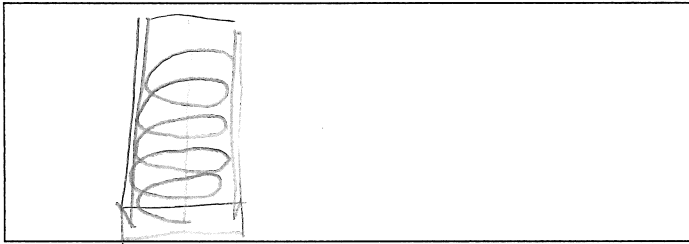
Make a free standing structure that takes the marble the longest time to go down

2-3. What Should We Think About?

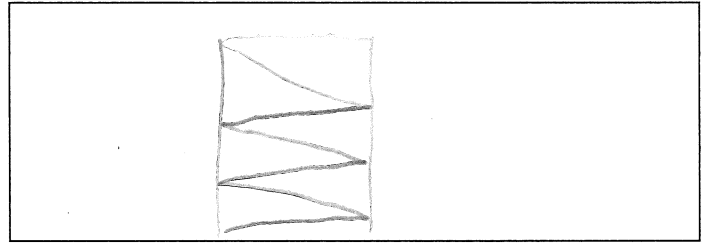
Before building, write or sketch a few things you think could affect travel time and reliable motion. What could help the marble keep moving? What could cause a run to fail?

- having good ramps - Smooth Tracks
- walls are a big factor - Steady supports

4. Brainstorm Idea A



4. Brainstorm Idea B



5. Choose a Solution

Which idea will you build first, and why do you think it is the better starting point?

I'm gonna try and Build A But if it doesn't work
then I'll Build B

6. Prototype Plan — Final Sketch Before Building

